CARE as Public Health Infrastructure

This analysis uses Seattle's 2025 911 CAD data to surface where CARE-style alternate response systems could most effectively complement SPD and **advance equitable access,** particularly in neighborhoods with **higher social vulnerability.** Seattle neighborhoods are identified where high volumes of CARE-eligible 911 calls coincide with **elevated Racial and Social Equity (RSE) scores**, and where alternative response units could be **prioritized** for **maximum impact** on police workload and community **health**.

The goal of this analysis is to provide a data-driven baseline that supports the continued expansion of the CARE model. By identifying where the system is currently excelling, it can aid development of a roadmap for replicating those successes in high-need areas.

After collapsing 546,208 call logs into 324,386 unique incidents in 2025 publicly available dataset, a transparent three-tier classification of calls identifies 20,867 events that seem to be eligible for CARE (noted as "clearly CARE Tier 2") events (i.e. 6.4% of all incidents). **Despite strong alignment in triage intent, 80.4% of these Tier 2 events were handled SPD-only, indicating room to expand CARE involvement *where operationally appropriate*.** Tier 2 burden generally rises with neighborhood need (**Seattle RSE Index; Spearman ρ≈0.40**), while a small set of "**destination**" neighborhoods (downtown/commercial/industrial) drives extreme per-capita rates. A neighborhood strategy matrix (Tier 2 demand × CARE involvement share), combined with **hourly response patterns**, highlights specific places and daytime windows where, in r**esource constrainted settings, targeted deployment experiments:** adjusting hours, staffing, placement, or dispatch rules- could meaningfully **extend** CARE reach while supporting SPD's core public safety mission.

Data Sources:

To build a reliable foundation, the raw data underwent a rigorous cleaning and "incident-first" transformation.

Call data: Seattle 911 CAD (2025), collapsed to one row per *cad_event_number* to avoid double-counting multi-unit activity.

Neighborhood vulnerability: Seattle *Racial and Social Equity (RSE) Composite Index* and sub-indices (Race/ELL/Origins, Socioeconomic disadvantage, Health disadvantage). The RSE index is designed to help identify where priority populations make up relatively large proportions of neighborhood residents, and includes sub-indices capturing race/ELL/origins, socioeconomic disadvantage, and health disadvantage. *[https://data.seattle.gov/dataset/Racial-and-Social-Equity-Composite-Index-Current/x5s4-2aie/about_data]*

Processing & Method overview:

1. Data Processing & Logical Deduplication

Raw CAD data is creates logical duplicates where one incident generates dozens of rows for every responding unit.

- **Raw Count:** 546,208 records.
- **Processed Count:** 324,386 unique events.
- **Logic:** Rows were collapsed by `cad_event_number`, ensuring that the retained record reflected the event's terminal classification and operational resolution.

E.g. of deduplication:

- -- Fresh Data Sanity Check --- total_rows=546208; unique_events=324386
- -- Inspection of CAD: *2025000353113* --- This was selected as the CAD with most duplicate rows for demo: Total Rows: 100

Columns that differ across these rows: ['call_sign_dispatch_id', 'call_sign_dispatch_time', 'call_sign_total_service_time_s_', 'call_sign_dispatch_delay_time_s_', 'call_sign_response_time_s_', 'call_sign_at_scene_time', 'call_sign_in_service_time', 'count_of_officers']

- **Confirmed:** Every row is physically unique. Duplication is logical (incident logs). The rows have differences in some columns (likely different units responding).
- Unique Incident IDs in Raw Data: 324386 Total Rows in Deduplicated File: 324386

2. Framework: CARE-Eligibility Tiering

To identify areas where alternative response can most effectively support the system, I developed a transparent three-tier classifier. This model uses regular expressions on `final_call_type` to categorize the 324k+ unique incidents.

- **Tier 0 – Traditional SPD:** Includes violent, property, or weapon-related categories. These are strictly excluded from CARE eligibility to prioritize officer and public safety.
- **Tier 1 – Potential CARE:** "Social maintenance" and nuisance calls. CARE may be highly relevant here, though these calls often occupy a "boundary category" where they may not yet be appropriate as a primary responder.
- **Tier 2 – Clearly CARE:** High-acuity behavioral health, welfare checks, and overdoses. These are categories where a non-punitive, human-centered response is plausibly the primary need.

Defining the "Care-Eligible" Lens

A CAD event is defined as care-eligible when the underlying need is **human-centered** and unlikely to be fully resolved by a single dispatch alone.

Key Philosophy: Emergency response solves the moment; care addresses the trajectory.

Note: this framework defines **eligibility**, not **appropriateness**.

- **Eligibility:** Indicates that a care-based intervention could plausibly assist in the long-term resolution of the incident.
- **Operational Appropriateness:** Recognizes that many Tier 1 and Tier 2 events may still require SPD presence for safety, legal requirements, or immediate scene stabilization.

Why this distinction matters: By focusing on **eligibility** (where care *could* help) rather than **appropriateness**, this framework remains grounded in operational reality. It is designed as a proactive tool to identify **where expanded coverage, staffing, or co-response experiments** could yield the highest marginal benefit, ensuring the data supports care team's resource needs.

An Iterative Approach:

This first-pass classification is a starting point. I**terating on these tiers with program and dispatch expertise** to ensure the definitions capture the on-the-ground reality of Seattle's streets would be the next right thing to do. Adjusting exclusions and validating "edge-case" call types with those who handle them daily will be essential for the next phase of this work.

3. Neighborhood-Level Equity Alignment (RSE Index)

To compare 911 demand with the City's equity priorities, Seattle's **Racial and Social Equity (RSE) Index** is realigned from the census-tract (GEOID) level to **Community Reporting Areas (CRAs)**. This spatial transformation ensures that equity scores are operationally compatible with dispatch reporting geographies.

Index Structure: The RSE Index is a composite "index of indices" from the Office of Planning & Community Development (OPCD). It aggregates indicators across three core domains: **Race/ELL/Origins**, **Socioeconomic Disadvantage**, and **Health Disadvantage**.

Population-Weighted Aggregation: For neighborhoods spanning multiple census tracts, scores were recomputed as population-weighted averages. This preserves the relative contribution of larger tracts and ensures the scores reflect the lived distribution of residents within dispatch zones.

Strategic Utility: This re-alignment allows for a direct, stratified analysis of **Tier 2 call burden** across equity quintiles. It serves as a spatial tool to identify where crisis demand intersects with the City's existing definitions of vulnerability, supporting place-based deployment and resource evaluation.

Systemwide results (2025):

1. 2025 System Baseline: Current Response & Opportunities

This baseline assessment establishes the current distribution of 911 calls and identifies where alternative response models can offer the most significant support to the existing system.

Current Response Distribution (2025): The vast majority of calls were handled by SPD (97.8%), with specialized CARE-only or Co-Response models currently presenting as an emerging segment of the city's emergency response (~2.11% combined).

Response Type	Frequency	Percentage
SPD Only	317,556	97.89%
CARE Only	3,847	1.19%
SPD/CARE Co-Response	2,983	0.92%
TOTAL	324,386	100%

Tiered Classification Profile: By applying the three-tier classification of **final call types**, we can see the total volume of incidents that fall within the "care-eligible" lens below:

Care Tier	Total Calls	Percentage
Tier 0 - Traditional SPD	177,837	54.80%
Tier 1 - Potential CARE	125,682	38.70%
Tier 2 - Clearly CARE	20,867	6.40%
TOTAL	324,386	100%

Together, these provide a demand segmentation view that supports **CARE expansion** planning and evaluation over time, while recognizing that eligibility is not the same as operational appropriateness.

2. Response Alignment:

Cross-tabulating **response category × care tier** (percentages **within each response category**) shows that CARE is generally being routed in the intended direction, with a few operationally informative edge patterns:

Response category	Tier 0 - traditional SPD response eligible: call type	Tier 1 - care-potential call type	Tier 2 - clearly care eligible call types	Total
CARE- responded	709 (18.4%)	51 (1.3%)	3,087 (80.2%)	3,847
SPD responded	176,658 (55.6%)	124,118 (39.1%)	16,780 (5.3%)	317,556
Co-response	470 (15.8%)	1,513 (50.7%)	1,000 (33.5%)	2,983

Response routing aligns strongly with Tier 2- care eligible call criteria, and Tier 1 appears to be the primary “co-response boundary space”- as intended.

- **High-Precision Targeting: 80.2% of CARE-only responses** land in Tier 2 ("CARE eligible"). **This confirms the program is successfully reaching the specific behavioral health incidents it was designed to prioritize.**
- **Co-Response as a Safety Bridge:** Half (**50.7%**) of all co-response events occur in Tier 1. This suggests the system effectively utilizes the partnership between officers and clinicians in "uncertain risk" scenarios.

Incident Evolution & Support: The **18.4%** overlap with Tier 0 - SPD only *call type* receiving **care-only response** likely reflects calls that evolve on-scene (e.g., a welfare check discovering a property crime) or instances where CARE provides trauma support on police calls. **This area is a priority for further collaborative investigation,** including narrative reviews, to ensure data accurately capture these complex operational nuances and safety of care-responders.

Investigating 18.4% of the incident highlighted red (falling under **traditional Spd response call type but getting CARE response**),

following is the distribution of the activity type-

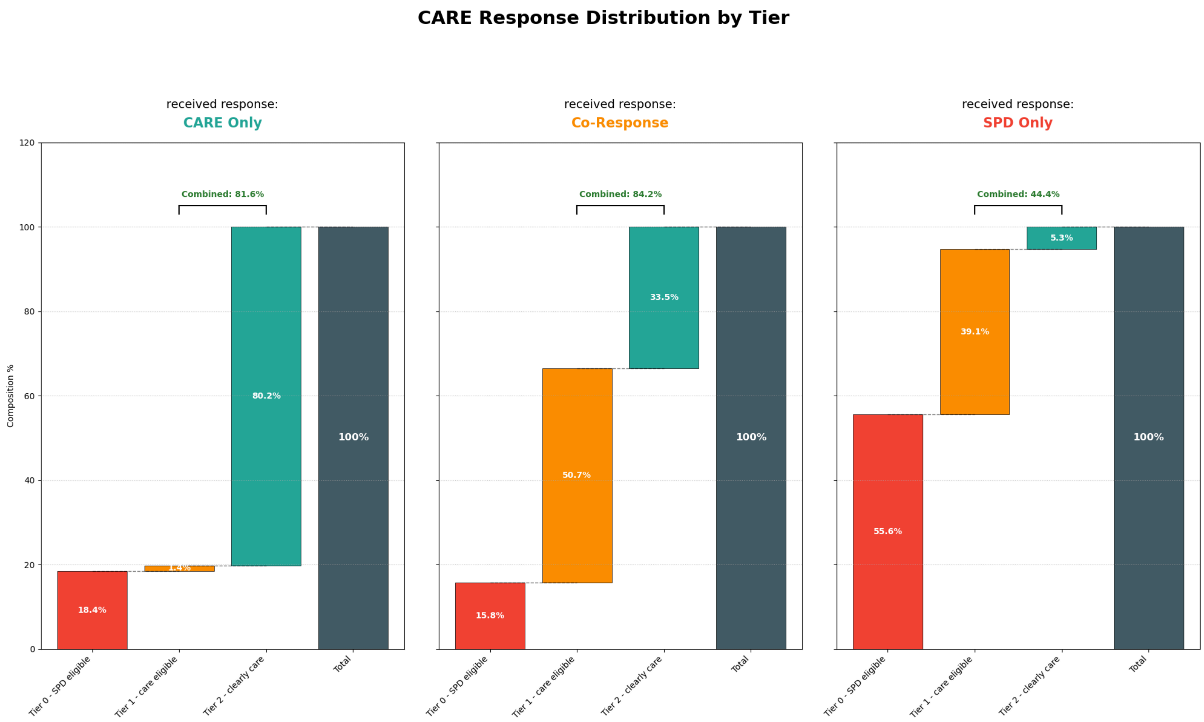
Activity Type	Count	
PREMISE CHECK, OFFICER INITIATED ONVIEW ONLY	420	59.2%
ASSIST PUBLIC - NO WELFARE CHK OR DV ORDER SERVICE	130	18.3%
DIRECTED PATROL ACTIVITY	93	13.1%
PREMISE CHECKS - CRIME PREVENTION	30	4.2%
ASSIST OTHER AGENCY - ROUTINE SERVICE	9	1.3%
TEST CALL ONLY	7	1.0%
FOLLOW UP	6	0.8%
ASSIST OTHER AGENCY - CITY AGENCY	3	0.4%
ASSIGNED DUTY - COMMUNITY,SCHOOL,SPECIAL EVENT	2	0.3%
MVC - COLLISION (ALL)	2	0.3%
PREMISE CHECKS - LICENSE INSPECTIONS	1	0.1%
BROADCAST AND CLEAR BY RADIO	1	0.1%
ASSIGNED DUTY - MEET W/ SUPERVISOR (OUT OF SVC)	1	0.1%
ASSIGNED DUTY - DETAIL BY SUPERVISOR	1	0.1%
ASSIGNED DUTY - OTHER ESCORT	1	0.1%
PREMISE CHECKS - BUSINESS CHECK	1	0.1%
GAS/MAINTENANCE/WASH/GARAGE	1	0.1%
sum:	709	1

Checking CAD incidents where care tier was SPD related but received CARE response.

This gives us more clarity:

CARE engagement on Tier 0 calls is primarily triggered during officer-initiated, routine activities (e.g., premise checks, directed patrols, public assists), suggesting CARE is being used as on-scene support when unmet care needs surface during everyday policing; not due to dispatch misclassification.

- **Operational reality:** CARE is functioning as a **field-based safety net**, activated when officers encounter behavioral health or social needs during routine patrol, not just through 911 triage.
- **Design implication:** Improving impact may depend less on changing dispatch rules and more on **streamlining SPD → CARE handoffs**, co-location, or rapid consult pathways during ONVIEW encounters.



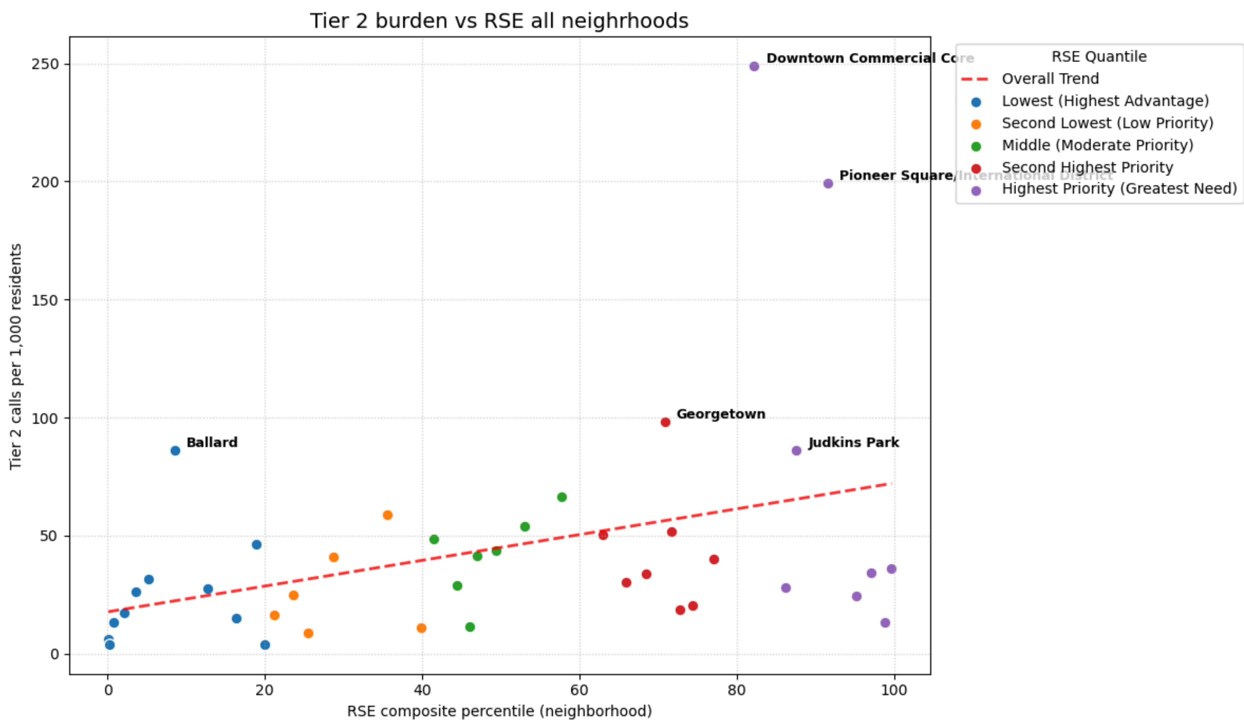
3. Crisis Burden & Neighborhood Equity:

This analysis examines how the demand for "Clearly CARE" (Tier 2) services intersects with neighborhood vulnerability, using the City's **Racial and Social Equity (RSE) Index**. The data suggests a clear relationship between social vulnerability and the frequency of crisis-related calls.

RSE Priority Category	Rate per 1,000 Residents
Lowest (Highest Advantage)	22.81
Second Lowest	29.87
Middle (Moderate Priority)	43.31
Second Highest Priority	34.48
Highest Priority (Greatest Need)	**65.72**

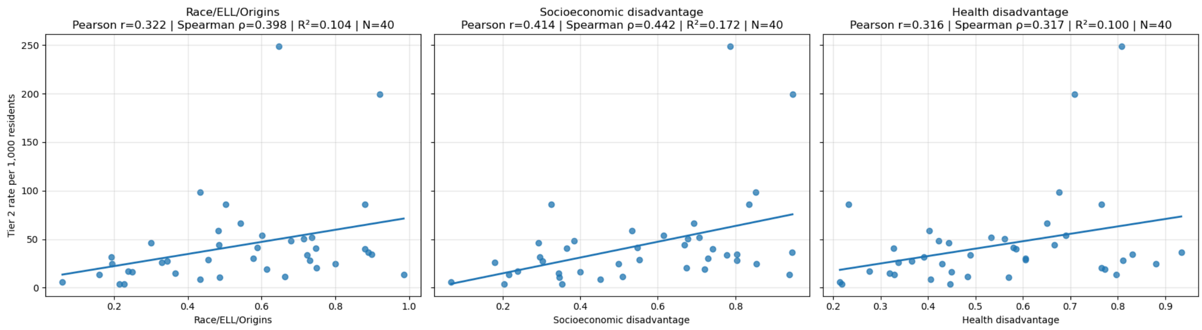
column 1: equity score; column2: tier-2 care-eligible call rate per 1000 resident.

- **The 3x Burden:** Neighborhoods in the **Highest Priority** category experience a crisis call rate of **65.72 per 1,000 residents-** nearly three times the call rate of the city's most advantaged areas.
- **Significant Positive Correlation:** There is a moderate, statistically significant relationship between neighborhood vulnerability and Tier 2 burden. (**Spearman's ρ=0.40**; **p=0.010**). This confirms that as RSE scores increase, the demand for care-based response typically follows.



3.1. Vulnerability Sub-Index Drivers

While the RSE sub-indices **(1) Race/ELL/Origins (2) Socioeconomic disadvantage (3) Health disadvantage are *deeply interconnected***, Socioeconomic Disadvantage is the most consistent predictor of Tier 2 crisis rates (spearman rho = 0.442).



Beyond the Index: The variance suggests that crisis volume is also shaped by environmental factors not captured by census-level vulnerability alone like **land use, service density and transit hubs**. This data confirms that socioeconomic need is a core driver, but reinforces that a successful deployment strategy must also account for the physical "hubs" where people naturally congregate and seek services. This supports a **place-based approach** to unit placement in addition to the citywide equity lens

3.2. Strategic Outliers & The "Destination Effect"

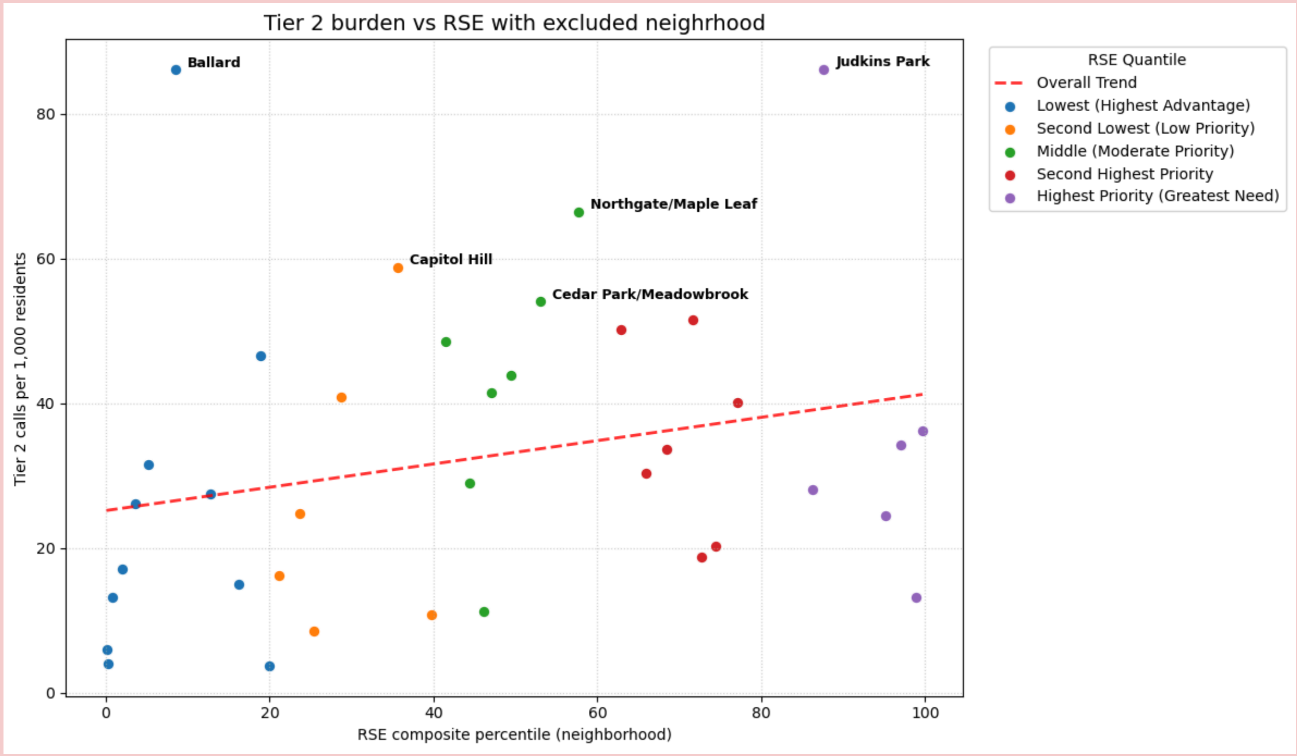
While neighborhood vulnerability is a key driver, several "Destination Neighborhoods" act as extreme outliers. These commercial and industrial hubs experience per-capita Tier 2 burdens far exceeding residential averages, likely due to high daytime visitor traffic relative to permanent resident counts.

Excluded Hubs: *Downtown Commercial Core, Pioneer Square/International District, and Georgetown.*

Excluding these hubs reveals a more moderate relationship between vulnerability and crisis rates (Spearman’s $\rho=0.33$). This suggests that an effective scaling strategy should be two-pronged:

1. **Equity-Based Placement:** Targeting high-priority residential zones (per RSE Index).
2. **Activity-Based Placement:** Targeting high-traffic destination hubs (e.g., Downtown or Ballard) where volume is driven by visitors and commercial density.

As expected, the outliers are not "noise" but operational realities. Identifying the "Destination Effect" helps us support a resource plan that balances citywide **social equity goals** with **operational efficiency** by placing units where the calls are happening, regardless of resident density.



This image excludes the top destination neighborhoods with tier 2 call burdens: downtown commercial core, pioneer square/international district & georgetown.

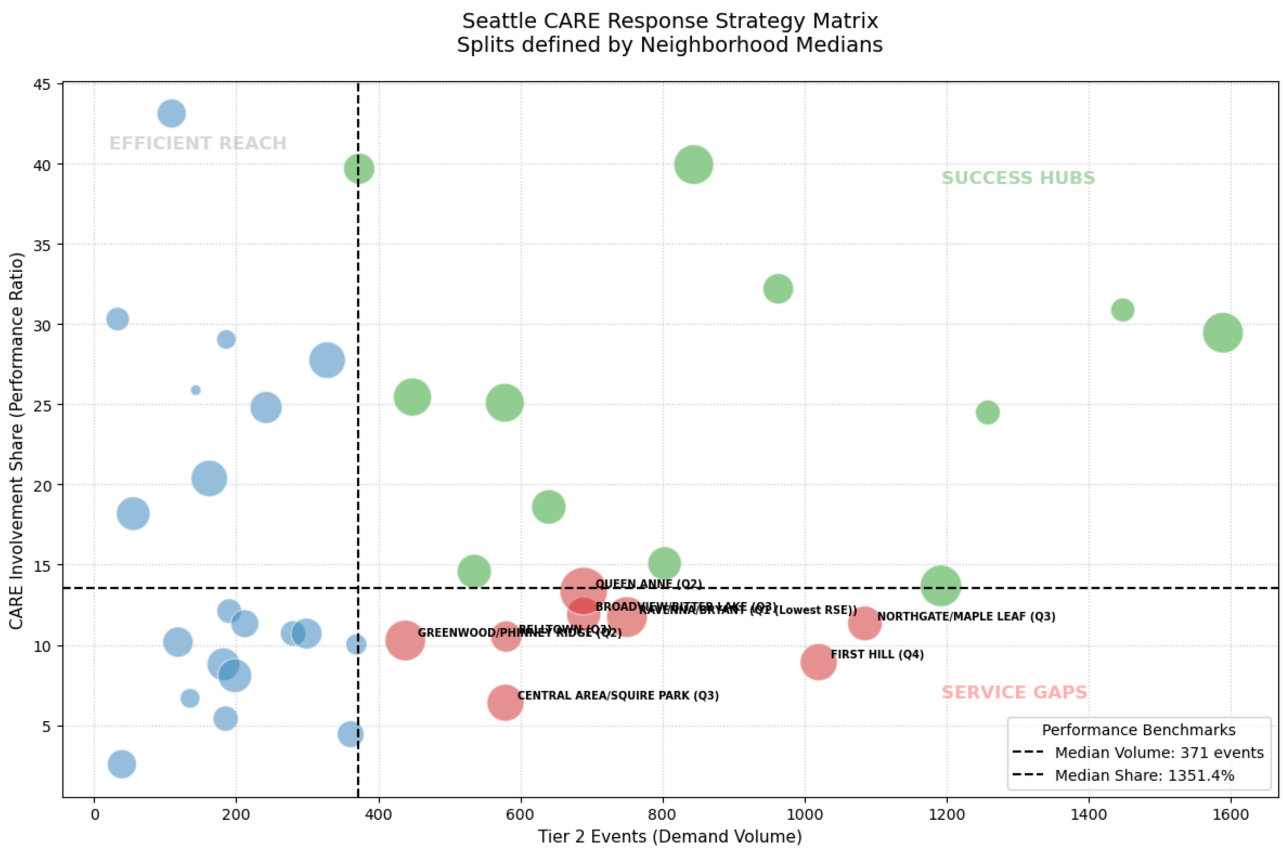
4. Neighborhood Strategy Quadrants

This analysis identifies how "Clearly CARE" (Tier 2) demand is currently met across different neighborhoods. By plotting **Tier 2 Volume** against **CARE Involvement Share**, we can visualize the system's successes and identify opportunities for strategic resource expansion.

Heterogeneity in Service Delivery:

The data shows a wide variance in how CARE resources are currently utilized:

- **Success Hubs:** Neighborhoods like **Capitol Hill (29%)**, **Ballard (32%)**, and **U-District (40%)** demonstrate comparatively strong integration where the CARE model is frequently and successfully deployed.
- **Service Gaps:** High-demand areas like **First Hill (8.9% share, N=1,020)** and **Central Area/Squire Park (6.4% share, N=579)** rely primarily on SPD-only responses for crisis calls.



Each point is a neighborhood. X-axis: Tier 2 demand volume. Y-axis: share of Tier 2 events with CARE involvement (CARE-only + co-response). Bubble size is scaled by neighborhood population. The upper-right quadrant indicates *high demand + high CARE involvement* ("success hubs"), while the lower-right quadrant indicates *high demand + low CARE involvement*.

Using this data-driven lens, we can pinpoint specific areas where increasing CARE capacity or refining local placement could offer the **highest marginal benefit** to the system.
Identified Service Gaps: First Hill, Northgate/Maple Leaf, Central Area/Squire Park, Belltown, Broadview/Bitter Lake, Queen Anne, Ravenna/Bryant, and Greenwood/Phinney Ridge.
Strategic Takeaway: First Hill stands out as a primary candidate for a targeted capacity pilot. Despite having one of the city's largest volumes of Tier 2 incidents, its CARE involvement share remains significantly below other high-demand zones. This data provides a robust, evidence-based baseline for exploring how expanded unit availability in this corridor could directly support SPD and meet care needs.

Interpreting "Success Hubs" vs. Unmet Tier 2 Volume:

1. The "Success Hubs" designation reflects **relative performance** (CARE involvement share), while unmet Tier 2 volume reflects **absolute system load**.
2. **High-performing neighborhoods can still have high unmet need.**

- Areas like **Capitol Hill** combine **strong CARE integration** with **exceptionally high Tier 2 demand** . As a result, even with higher-than-average CARE involvement, the remaining unmet volume is large in absolute terms.

dispatch_neighborhood	tier2_volume	care_share	unmet_tier2_volume
CAPITOL HILL	1589	0.294525	1121.0
DOWNTOWN COMMERCIAL	1448	0.308702	1001.0
SLU/CASCADE	1133	0.139453	975.0
NORTHGATE	1085	0.113364	962.0
FIRST HILL	1020	0.089216	929.0

Operational implication: capacity pressure vs. integration gaps.

1. **High demand + high CARE share** (e.g., Capitol Hill, Downtown): indicate **sustained pressure** and potential benefit from *additional capacity*.
2. **High demand + low CARE share** (e.g., First Hill, Northgate): indicate **service gaps** where targeted placement or pilots could yield high marginal gains.

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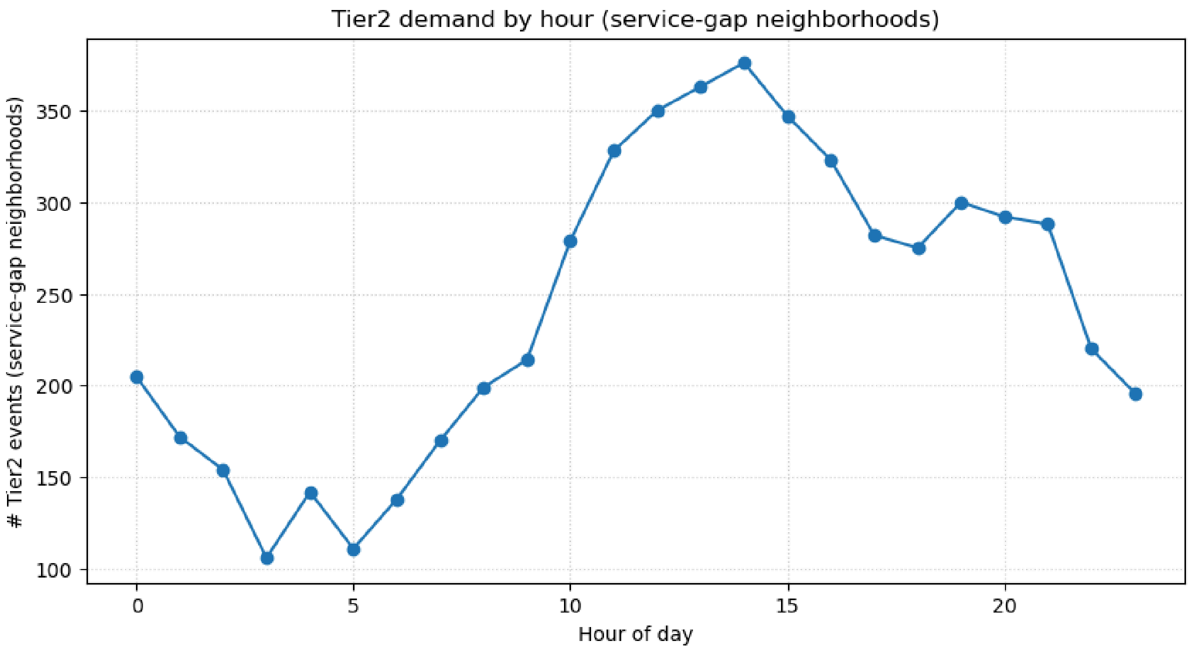
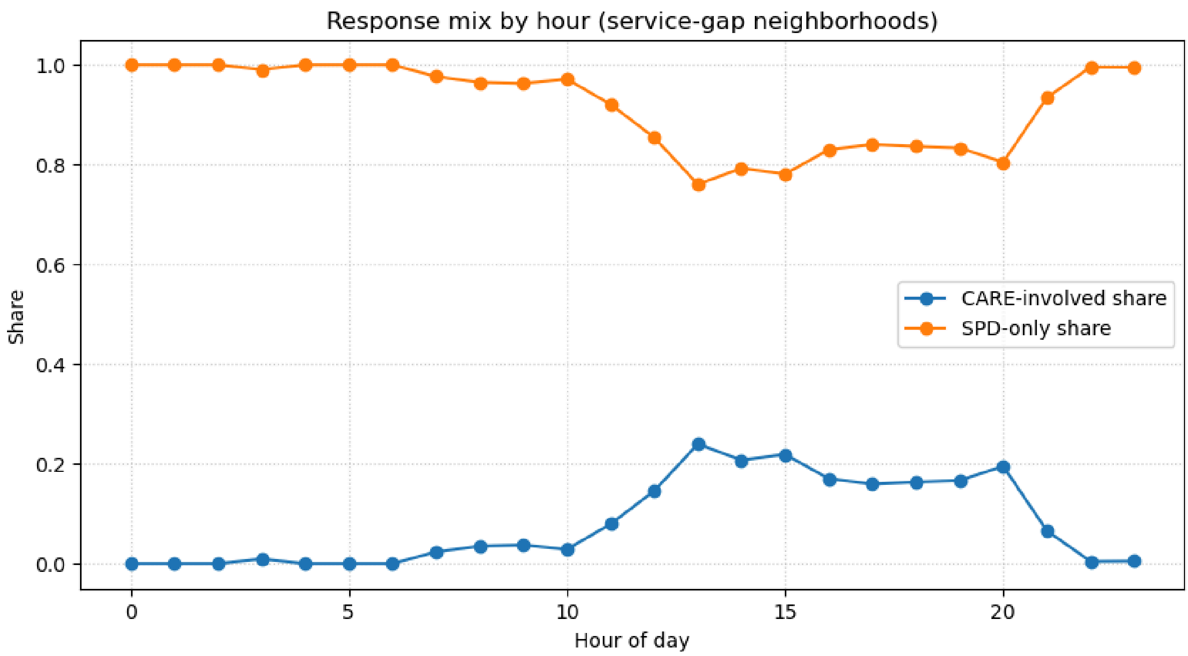
  To determine if temporal factors drive the difference between neighborhood types, I compared the hourly CARE involvement in neighbourhoods of **Success Hubs** versus **Service Gaps**.

  1. **Daytime Divergence:** The primary difference emerges during active hours (**12:00–20:00**). While both groups see a rise in call volume, **Success Hubs** achieve significantly higher CARE involvement shares than **Service Gaps** during this window.



Tier2 demand by hour (service-gap neighborhoods)
Tier2 events (service-gap neighborhoods)
100
150
200
250
300
350
0
5
10
15
20
Hour of day



Response mix by hour (service-gap neighborhoods)
Share
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
Hour of day
CARE-involved share
SPD-only share

	Service-gap neighborhood			Success neighborhood		
Hour	**# events**	**CARE share**	**SPD-only share**	**# events**	**Care Share**	**SPD Only Share**
0	205	0.00%	100.00%	279	0.00%	100.00%
1	172	0.00%	100.00%	251	0.40%	99.60%
2	154	0.00%	100.00%	227	0.88%	99.12%
3	106	0.94%	99.06%	174	0.00%	100.00%
4	142	0.00%	100.00%	170	0.00%	100.00%
5	111	0.00%	100.00%	165	0.00%	100.00%
6	138	0.00%	100.00%	208	0.48%	99.52%
7	170	2.35%	97.65%	235	2.13%	97.87%
8	199	3.52%	96.48%	281	0.36%	99.64%
9	214	3.74%	96.26%	401	1.99%	98.01%
10	279	2.87%	97.13%	432	3.24%	96.76%
11	328	7.93%	92.07%	482	6.22%	93.78%
12	350	14.57%	85.43%	603	25.87%	74.13%
13	363	23.97%	76.03%	888	48.76%	51.24%
14	376	20.74%	79.26%	879	50.40%	49.60%
15	347	21.90%	78.10%	735	42.45%	57.55%
16	323	17.03%	82.97%	678	38.79%	61.21%
17	282	15.96%	84.04%	656	39.63%	60.37%
18	275	16.36%	83.64%	621	44.93%	55.07%
19	300	16.67%	83.33%	647	43.74%	56.26%
20	292	19.52%	80.48%	564	37.94%	62.06%
21	288	6.60%	93.40%	421	12.35%	87.65%
22	220	0.45%	99.55%	359	0.28%	99.72%
23	196	0.51%	99.49%	315	0.00%	100.00%

Strategic Takeaway: This confirms that closing the gap in neighborhoods like First Hill or Northgate isn't necessarily about changing *when* we work, but rather **where we are based** and **how many units** are available in those specific sectors.

- ## 5. Field Discovery, Equity, and System Design

  CARE is partly field-discovered: Approximately **21% of Tier 2 (“Clearly CARE”) crises are identified through officer patrol (ONVIEW)** rather than through 911 dispatch, indicating that CARE operates as a field-responsive system in addition to a dispatch-routed one.
  This points to:

Call Type	Count	Percentage	Distribution
911	**11,921**	57.1%	
Telephone Other, Not 911	**4,539**	21.8%	
On-View	**4,363**	20.9%	
Text Message	**32**	0.2%	
Alarm Call (Not Police Alarm)	**7**	0.03%	
Proactive (Officer Initiated)	**3**	0.01%	
In Person Complaint	**1**	0.00%	
Scheduled Event (Recurring)	**1**	0.00%	
Total	**20,867**	**100.0%**	

Call Type distribution for tier 2 call care eligible dataset.
The "Pre-Crisis" Window: Onview incidents are often officers noticing someone in distress *before* it escalates to a 911 call from a bystander.

1. This represents a **critical pre-crisis intervention window**, where:
   - care-based response may prevent escalation,
   - harm reduction and stabilization can occur earlier,
   - long-term public health outcomes may be improved.
2. This positions CARE not only as an alternative response, but as a **preventative public health mechanism**.

Geographic implications for equity: If care needs are disproportionately discovered through patrol, **"quieter" or lower-visibility neighborhoods may face barriers to CARE access**, raising an important equity question about how care is distributed across the city.

Visibility may shape access: From a health equity and social determinants lens, this suggests that **visibility in public space can influence who receives**

care-based response, as neighborhoods with higher proactive patrol presence or greater foot traffic are more likely to surface care needs through observation.

These findings highlight that scaling CARE equitably requires attention not only to dispatch logic, but also to field-level workflows and geographic patterns of patrol presence.

#	Dispatch Neighborhood	Tier2 Events Check	Tier2 OnView Share	Tier2 911 Share	Tier2 Received Care Share	N Tier2 OnView	N 911 OnView	N Received Care
8	CAPITOL HILL	1589	44.81%	37.76%	29.45%	712	600	468
14	DOWNTOWN COMMERCIAL	1448	34.88%	49.45%	30.87%	505	716	447
51	SLU/CASCADE	1133	21.18%	54.28%	13.95%	240	615	158
41	NORTHGATE	1085	12.07%	66.36%	11.34%	131	720	123
18	FIRST HILL	1020	6.76%	67.45%	8.92%	69	688	91
56	UNIVERSITY	844	34.95%	42.30%	39.93%	295	357	337
27	LAKECITY	803	13.20%	67.12%	15.07%	106	539	121
10	CHINATOWN/INTERNATIONAL DISTRICT	769	24.45%	58.26%	20.55%	188	448	158
48	ROOSEVELT/RAVENNA	750	8.67%	72.53%	11.73%	65	544	88
4	BALLARD SOUTH	689	38.46%	42.82%	42.09%	265	295	290
6	BITTERLAKE	689	8.85%	65.75%	11.90%	61	453	82
45	QUEEN ANNE	689	9.87%	64.59%	13.35%	68	445	92
5	BELLTOWN	580	10.52%	68.10%	10.52%	61	395	61
9	CENTRAL AREA/SQUIRE PARK	579	4.66%	63.90%	6.39%	27	370	37
57	UNKNOWN	534	43.82%	18.54%	26.03%	234	99	139
44	PIONEER SQUARE	489	29.86%	53.78%	30.67%	146	263	150
39	NORTH BEACON HILL	485	23.92%	62.27%	22.68%	116	302	110
22	GREENWOOD	385	5.45%	74.03%	9.09%	21	285	35
26	JUDKINS PARK/NORTH BEACON HILL	369	4.34%	74.25%	10.03%	16	274	37
33	MILLER PARK	361	2.49%	77.84%	4.43%	9	281	16
1	ALASKA JUNCTION	334	35.93%	43.71%	35.93%	120	146	120
19	FREMONT	328	27.13%	43.29%	27.74%	89	142	91
50	SANDPOINT	299	9.36%	55.85%	10.70%	28	167	32
36	MOUNT BAKER	280	8.93%	71.79%	10.71%	25	201	30
3	BALLARD NORTH	274	7.30%	67.15%	7.30%	20	184	20
12	COLUMBIA CITY	233	42.49%	44.21%	43.78%	99	103	102
46	RAINIER BEACH	216	27.31%	57.41%	26.85%	59	124	58
23	HIGH POINT	212	19.81%	53.30%	11.32%	42	113	24
35	MORGAN	212	6.60%	65.57%	8.49%	14	139	18
49	ROXHILL/WESTWOOD/ARBOR HEIGHTS	209	27.27%	50.24%	26.79%	57	105	56
2	ALKI	208	56.25%	33.65%	52.88%	117	70	110
7	BRIGHTON/DUNLAP	201	9.95%	68.16%	9.45%	20	137	19
24	HIGHLAND PARK	185	3.24%	70.27%	5.41%	6	130	10
31	MAGNOLIA	182	2.75%	68.68%	8.79%	5	125	16
40	NORTH DELRIDGE	181	4.42%	67.96%	12.71%	8	123	23
52	SODO	177	30.51%	53.11%	29.38%	54	94	52
38	NORTH ADMIRAL	165	21.82%	56.36%	23.03%	36	93	38
58	WALLINGFORD	162	15.43%	61.11%	20.37%	25	99	33
32	MID BEACON HILL	155	5.81%	64.52%	5.81%	9	100	9
11	CLAREMONT/RAINIER VISTA	151	5.30%	68.21%	5.30%	8	103	8
21	GEORGETOWN	143	27.97%	46.15%	25.87%	40	66	37
55	SOUTH PARK	135	3.70%	68.15%	6.67%	5	92	9
30	MADRONA/LESCHI	118	5.93%	74.58%	10.17%	7	88	12
47	RAINIER VIEW	118	4.24%	72.88%	0.85%	5	86	1
28	LAKEWOOD/SEWARD PARK	109	44.04%	37.61%	43.12%	48	41	47
37	NEW HOLLY	107	10.28%	59.81%	9.35%	11	64	10
53	SOUTH BEACON HILL	91	27.47%	56.04%	6.59%	25	51	6
25	HILLMAN CITY	64	6.25%	60.94%	6.25%	4	39	4
17	FAUNTLEROY SW	55	14.55%	63.64%	18.18%	8	35	10
42	PHINNEY RIDGE	53	7.55%	67.92%	18.87%	4	36	10
34	MONTLAKE/PORTAGE BAY	39	2.56%	58.97%	2.56%	1	23	1
16	EASTLAKE - WEST	35	2.86%	62.86%	8.57%	1	22	3
29	MADISON PARK	33	24.24%	51.52%	30.30%	8	17	10
54	SOUTH DELRIDGE	33	15.15%	63.64%	12.12%	5	21	4
20	GENESEE	32	12.50%	75.00%	21.88%	4	24	7
15	EASTLAKE - EAST	24	8.33%	70.83%	8.33%	2	17	2
13	COMMERCIAL DUWAMISH	9	22.22%	66.67%	22.22%	2	6	2
43	PIGEON POINT	9	11.11%	66.67%	0.00%	1	6	0
0	-	6	100.00%	0.00%	50.00%	6	0	3
					sum check ok:	4363	11921	

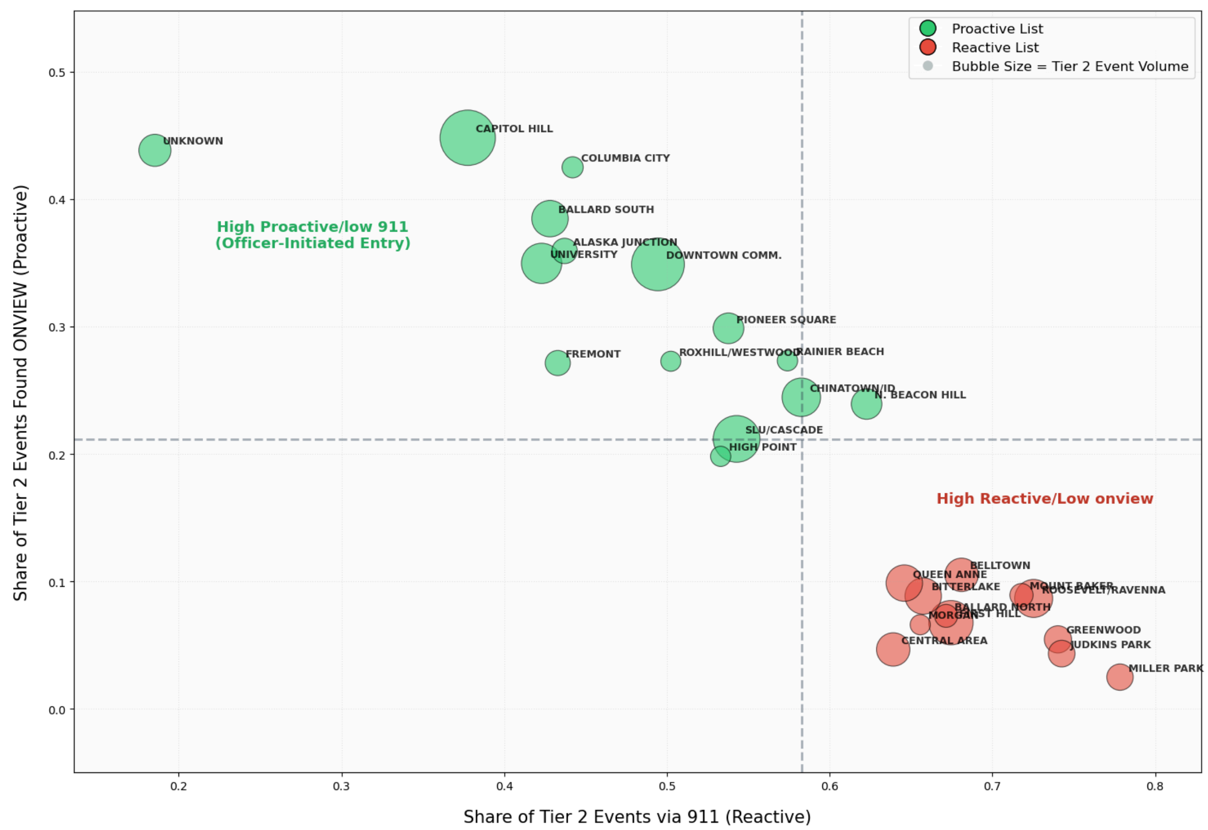
First Hill has high RSE score (Q4).

To avoid over-interpreting small numbers, I restricted this analysis to neighborhoods with high Tier 2 volume, so we're comparing places where crisis demand is meaningfully present.

3. **Reactive Zones (high 911, low ONVIEW):** crisis need shows up primarily through **calls for service**, not patrol discovery
4. **Proactive Zones (low 911, high ONVIEW):** crisis discovery is disproportionately **patrol-mediated** (visibility effect).
    - **Reactive Zones (High 911, Low ONVIEW):** Neighborhoods such as **First Hill, Lake City, Roosevelt/Ravenna, Bitter Lake, Queen Anne, and Belltown** show high reliance on **911 calls** to surface Tier 2 crises, with relatively low patrol-based discovery. In these areas, **CARE access is primarily mediated through dispatch**, suggesting that crisis visibility depends more on bystander reporting than field observation.
    - **High demand does not imply proactive discovery:** Several Reactive Zones have substantial Tier 2 volumes (e.g.,**First Hill: N=1,020; Lake City: =803**), yet only a small fraction of Tier 2 incidents are discovered ONVIEW (Tier 2 ONVIEW share often below 10%). This indicates **limited opportunity for early, pre-crisis intervention** through patrol presence alone.

Proactive Zones tend to be areas with **high foot traffic, mixed-use land patterns, or service density**, where distress is more visible in public space. This reinforces the idea that **geography and visibility shape how care needs are surfaced**, not just underlying need.

CARE Entry points by neighbourhood for Tier 2 care eligible cads



Summary

This analysis provides an evidence-based framework to optimize Seattle's crisis response, demonstrating that while the **CARE** model is highly effective, strategic geographic expansion is the key to closing current service gaps.

- **Operational Precision:** Currently ~0.9% of the 911 calls receive care response though we observe ~6.4% of the calls to be care-eligible. The tiered model is work in progress and needs to go through iterations for more refined results; we do see **80.2%** of CARE-only responses are accurately targeted to "Clearly CARE" (Tier 2) incidents. Co-response serves as a vital safety bridge, handling **50.7%** of "Potential CARE" (Tier 1) cases.
- **The Equity-Demand Link:** Care-eligible 911 Call Crisis burden is not distributed equally. High-priority (RSE) neighborhoods face a **3x higher crisis rate** than advantaged areas, with socioeconomic disadvantage serving as the primary predictor ($\rho=0.44$) compared to Health disadvantage subindex or Race/Origin/ELL subindex. Though all three subindices are highly interconnected and show positive correlation.
- **Identifying the Service Gap:** "Success Hub" neighborhood where the tier 2 care eligible burden is high and the share of CARE response to these events are also high like **Capitol Hill (29% share)** show strong integration of care response system. However, several high-demand areas, most notably **First Hill,** have persistently low CARE involvement despite large Tier 2 care eligible call volumes. Differences are driven less by time of day, suggesting that targeted geographic expansion could yield high marginal benefits.
- **A Two-Pronged Scaling Strategy:** To maximize system relief for SPD and improve health equity, expansion should prioritize a dual approach:
    1. **Equity-Based Placement** in high-vulnerability residential zones.
    2. **Activity-Based Placement** in high-traffic "Destination Hubs" (e.g., Downtown, First Hill, Ballard).

Strategic Takeaway: The data confirms that CARE is landing in the right places with high precision. By decentralizing units into identified "Service Gaps" like **First Hill**, the city can immediately increase the marginal benefit of the program stabilizing more scenes with care-centered resources. This analysis shares data-driven evidence to scale CARE through **strategic, place-based investments** that balance social equity goals with operational efficiency.

- **Pre-Crisis & Health Equity Lens:** About **21% of Tier 2 crises are identified ONVIEW**, when officers encounter distress before escalation to a 911 call. From a public health perspective, this represents a key **pre-crisis prevention window**, where early care-based response can reduce escalation and improve downstream outcomes. However, because ONVIEW discovery depends on patrol presence and public visibility, **access to CARE may be shaped by environmental and structural factors rather than need alone**. High-activity areas are more likely to surface care

needs through observation, while **high-vulnerability neighborhoods like First Hill (high RSE score) rely disproportionately on reactive 911 pathways despite substantial Tier 2 demand**. This suggests that field-based discovery functions as a **structural determinant of access**, and that equitable CARE scaling must account for patrol geography and field-level workflows, not dispatch logic alone.

Strategic Takeaway: The data confirms that CARE is landing in the right places with high precision, but current access is heavily mediated by 911 dispatch. By identifying **"Reactive Zones"** like First Hill and Lake City where high Tier 2 volumes exist but "onview" discovery is **<10%** we uncover a massive opportunity for **preventative public health.** Scaling CARE through **strategic, place-based investments** can aim to close the **Visibility Gap.** Moving units into these high-demand/low-visibility areas allows the city to stabilize scenes *before* escalation, balancing social equity goals with a proactive, care-centered intervention model that doesn't rely solely on bystander reporting or police patrol.

Slides for easy understanding:
https://www.canva.com/design/DAG_7bdUcxM/ahWnDEz42NnXj76hJPXooQ/edit?utm_content=DAG_7bdUcxM&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

Lit Review (in progress)

Risk Managed Demand: Operational Risk Management in Police Response to Calls for Service - Loren T. Atherley, 2025